Master Software Testing with Java, Selenium & Real-Time Projects

Build your career in IT with real-time manual and automation testing skills

Expert Trainers
Learn from industry professionals

Java + Selenium
Core automation tech stack

Manual + Automation
Complete testing coverage

Real-Time Projects
Hands-on industry practice

Updated Curriculum
Latest tools and frameworks

Placement Support
Resume, mock interviews & referrals

🚀 **Join Our Next Batch — Start Your Testing Career Today!**
XpertsEdgeTechnologies · 8925175332 / 8870783300 · No. 2110A, 13th Main Road, Anna Nagar, Chennai - 600040

About XpertsEdgeTechnologies

XpertsEdgeTechnologies is a professional IT training institute focused on developing skilled software testing engineers ready for the modern tech industry. Our institute was built with a singular mission: to bridge the gap between academic knowledge and real-world IT demands. We understand that the software industry requires engineers who can hit the ground running — and that's exactly what our curriculum is designed to deliver.

At XpertsEdgeTechnologies, we don't just teach theory. Every module is designed around practical, hands-on tasks that simulate the challenges you will face on a real project. Our trainers bring their industry experience directly into the classroom, giving learners a competitive advantage from day one. Whether you are a fresh graduate, a career switcher, or someone moving from a non-IT background, our program meets you where you are and takes you to where the industry needs you to be.

Our training environment is built for focus, growth, and results. From air-conditioned classrooms to LED projector sessions, from weekly doubt-clearing sessions to one-on-one mock interviews, every element of our institute is optimized for your success. We measure our success by yours — and our community of placed QA engineers across the country is a testament to our commitment.

> **Mission:** To provide high-quality software testing training with real-time project exposure and help students build a successful IT career.

Practical Training

Every concept backed by real coding tasks and test scenarios

Real-Time Exposure

Work on live industry-style testing projects and automation tasks

Industry-Ready Skills

Graduate with the tools, frameworks, and confidence employers seek

Career Guidance

Resume prep, mock interviews, and job referral support

Why Choose XpertsEdgeTechnologies?

Choosing the right training institute can define the trajectory of your entire career. At XpertsEdgeTechnologies, we have carefully designed every aspect of the program to maximize your learning, your confidence, and your job-readiness. We are not a generic classroom — we are a career launchpad for software testing professionals.

Our trainers are not just educators — they are working professionals who have built real automation frameworks, managed QA teams, and solved real testing challenges in production environments. When they teach, they bring that context into every explanation, every example, and every exercise. You will learn not just the "what" but the "why" and the "how" — the insights that only experience can provide.

Industry-Experienced Trainers

Learn from professionals with real-time project experience and current IT knowledge. No outdated textbook content — only what the industry actually uses.

Complete Placement Support

Resume guidance, mock interviews, referral connections, and job preparation. We stay with you until you land your role.

Hands-On Project Experience

Real-time testing projects that mirror actual company environments. You will graduate with a portfolio, not just a certificate.

Latest Tools & Updated Syllabus

Selenium 4, TestNG, Cucumber, Jenkins, Git, Maven, SQL — all the tools modern QA engineers use daily.

Premium Classroom Environment

Air-conditioned classrooms, LED projector sessions, and offline learning with direct trainer interaction and real-time doubt support.

1-on-1 Mock Interviews

Expert-led personal mock interviews with detailed feedback to sharpen your communication and technical confidence before the real thing.

Software Testing Program Highlights

The Java Selenium Software Testing Program at XpertsEdgeTechnologies is engineered from the ground up to deliver measurable outcomes. Every module, every tool, and every project is aligned with what IT companies actually expect from their QA hires today. Here is what makes this program stand out from every other testing course in the market.

🎓 Expert Trainers

Learn from industry professionals with real-time project experience. Trainers bring current, company-level insights into every session — no outdated content, ever.

💼 Placement Support

Resume guidance, mock interviews, referrals, and comprehensive job preparation. We actively connect qualified graduates with hiring companies in the QA space.

📚 Updated Curriculum

Designed with the latest industry tools — manual testing fundamentals, automation frameworks, CI/CD integration, and API testing concepts all in one structured roadmap.

🖥️ Real-Time Practice

Work on practical testing scenarios, live automation tasks, and project-based learning that simulates what you will encounter in your first QA role.

🏫 Classroom Training

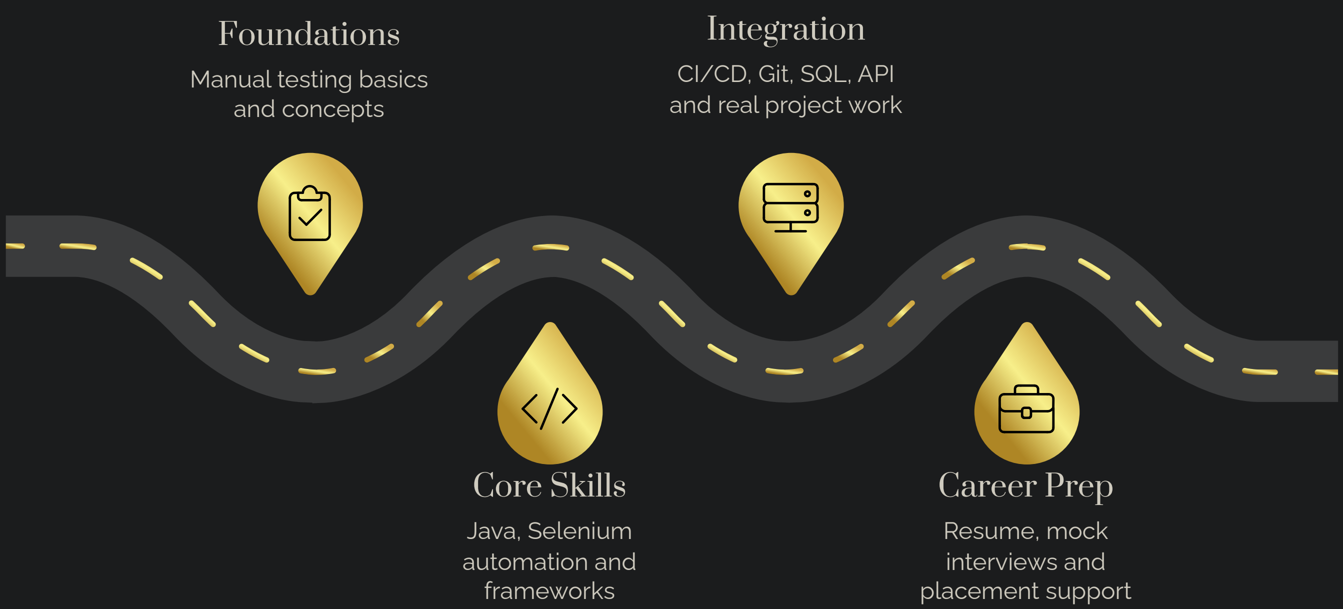
Offline learning with direct trainer interaction, LED projector sessions, and immediate doubt resolution. No pre-recorded lectures — only live, engaging training.

🎯 Interview Focus

1-on-1 mock interviews with expert feedback on technical answers, communication, and confidence. Practice until you are genuinely interview-ready.

Java Selenium Testing Roadmap

Our program follows a carefully sequenced roadmap that builds your skills layer by layer — from foundational manual testing concepts all the way to real-time project practice and career placement support. Each phase is designed to reinforce the previous one, ensuring you never feel lost or overwhelmed as you progress through the curriculum.



This eight-phase journey is structured so that every learner — whether they come from a technical background or not — can follow along and build genuine expertise. By the time you reach Phase 8, you will not just know the tools; you will know how to use them in a real project, explain your work in an interview, and present yourself as a confident, hire-ready QA professional.

8 Learning Phases From basics to advanced automation	**15+** Core Skills Tools, frameworks, and languages
100% Practical Focus Hands-on learning throughout	**1:1** Mock Interviews Personal expert feedback sessions

Java Module — Core for Testing

WEEK 1: BASICS & OOPS

Java is the backbone of modern Selenium automation testing. Before you can write a single line of Selenium code, you need to think like a Java developer — and that is exactly what this module builds. Over the course of Week 1, you will move from complete beginner to a confident Java programmer ready to apply OOP concepts in real automation scenarios.

The module begins with the fundamentals: setting up Eclipse, understanding Java project structure, working with packages, and writing your first programs. From there, you will progress through data types, wrapper classes, and the powerful autoboxing/unboxing mechanism that Java automation frameworks rely on extensively. By the end of Week 1, you will have a working understanding of all four pillars of object-oriented programming — inheritance, abstraction, polymorphism, and encapsulation — applied in the context of software testing.

Week 1 Topics

- Introduction, Installation & Project Setup using Eclipse
- Packages: Same and Different Package
- Data Types
- Wrapper Class
- Autoboxing and Unboxing
- Inheritance
- Abstraction
- Polymorphism

Learning Outcome

Students completing the Java Foundation module will have a thorough understanding of Java fundamentals required for Selenium automation testing. You will be able to set up a Java project from scratch, write object-oriented programs, and understand how these OOP principles translate directly into building clean, maintainable test automation code.

Every Java concept taught in this module is directly connected to a real Selenium use case — so nothing feels abstract or disconnected.

Advanced Java Concepts for Test Automation

Once you have mastered the Java fundamentals, this module takes you deeper into the advanced concepts that separate a beginner tester from a professional automation engineer. Every topic in this module is chosen because it appears frequently in real-world Selenium automation frameworks used by companies today.

Collections are one of the most important topics for any automation tester — you will use Lists, Sets, and Maps constantly when handling dynamic web elements, test data, and framework configurations. Exception handling ensures your automation scripts are robust and don't fail unexpectedly in production-like environments. The Singleton Pattern, in particular, is a fundamental design principle used in building the Base Class of every professional automation framework.

1

Collections

List, Set, Map — managing dynamic data in automation scripts

2

Exception Handling

Build robust, production-safe automation code

3

Singleton Pattern

Core design pattern for automation framework Base Class

4

File Handling

Read and write test data files for data-driven testing

5

Constructors & Chaining

Build flexible, reusable class structures for frameworks

6

Screenshot Concept

Capture test evidence programmatically during automation runs

Additional Topics Covered

- Sorting Programs
- Scanner Class
- Encapsulation

Outcome

Students will learn Java concepts used directly in real-time automation frameworks — preparing them to build, extend, and maintain professional-grade Selenium test suites with confidence.

Selenium Module — Automation Testing

WEEK 1: INTRODUCTION & LOCATORS

Selenium WebDriver is the gold standard for web application automation testing — and this module gives you a complete, hands-on foundation in the tool that powers QA automation worldwide. You will begin by understanding how Selenium is architected, how WebDriver communicates with browsers, and how to set up your first automated test from scratch.

Locators are the heart of Selenium automation. Without accurate locators, your automation scripts cannot interact with web elements — and inaccurate locators are one of the top reasons automation frameworks fail in real projects. This module dedicates significant time to mastering every type of locator: ID, Name, XPath, CSS Selector, Link Text, and TagName. You will learn not just how to use them, but when to use each one and how to write them correctly for complex, dynamic web pages.

Selenium Intro & Architecture

Understand how WebDriver communicates with browsers at the architectural level

WebDriver Methods

Core browser control methods: navigate, find, click, type, and verify

ID & Name Locators

The fastest and most reliable locators for web elements

XPath

Powerful path expressions for navigating complex DOM structures

CSS Selector

Performance-optimized locator strategy for modern web apps

Link Text & TagName

Specialized locators for anchor elements and HTML tag targeting

Outcome: Students will learn how to automate web applications using Selenium WebDriver — from the first browser launch to complex element interaction.

Core Selenium Automation Concepts

This module moves beyond basic element interaction and dives into the advanced Selenium techniques that real QA automation engineers use every single day. These are the concepts that distinguish a junior tester from a confident automation professional — and every topic here is pulled directly from real project requirements in the industry.

Handling alerts, dropdowns, frames, and multiple windows are challenges that every automation engineer encounters in live projects. Understanding waits — both implicit and explicit — is critical for writing stable, non-flaky automation scripts. The Actions Class and Robot Class enable you to simulate complex user interactions like drag-and-drop, keyboard shortcuts, and right-click menus that are impossible with basic WebElement methods alone.

Core Automation Topics

- Alerts Handling
- Debugging Techniques
- Implicit and Explicit Waits
- Actions Class
- Robot Class
- Frames Handling
- JavaScript Executor
- Screenshot Handling
- Dropdown Handling using Select Class
- Window Handling
- Web Table Handling

Why These Topics Matter

Each of these concepts directly addresses a real challenge you will encounter when automating modern web applications. Waits prevent test failures caused by slow page loads. JavaScript Executor handles elements that normal WebDriver methods cannot reach. Web Table Handling enables you to validate data-heavy enterprise applications — a very common requirement in real projects.

Outcome

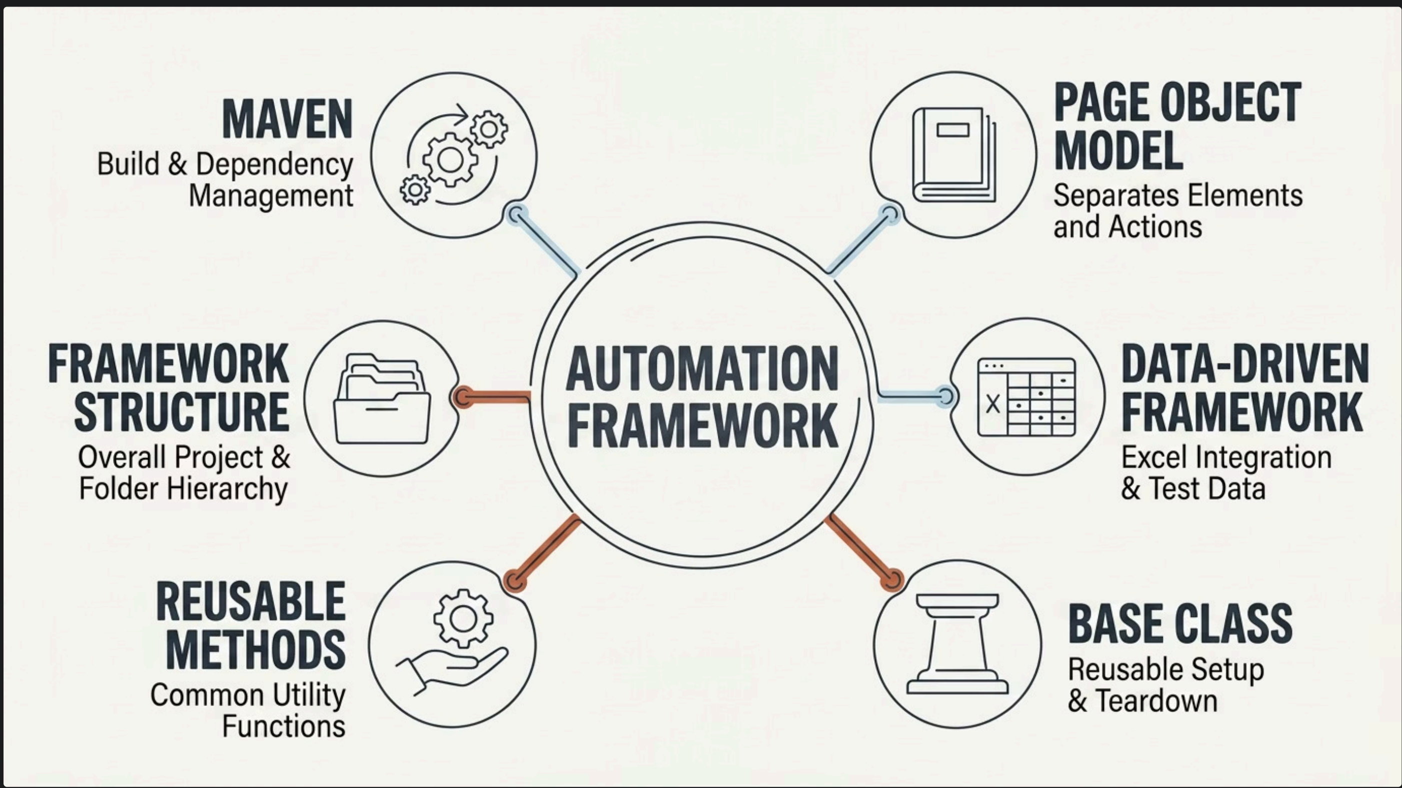
Students will be equipped to handle real-time web automation challenges confidently — writing scripts that are stable, reliable, and production-ready.

All topics are demonstrated with live browser automation tasks on real websites during class sessions.

Automation Framework Development

Building a framework is the hallmark of a senior automation engineer — and at XpertsEdgeTechnologies, we teach you how to build one from scratch. This module is one of the most valuable in the entire program because it is where all your Java, Selenium, and testing knowledge comes together into a professional, maintainable automation architecture.

The Page Object Model is the industry-standard design pattern for organizing Selenium test code. Instead of writing raw locators and actions scattered throughout test files, POM separates page-specific logic from test logic — making your codebase cleaner, more readable, and dramatically easier to maintain when the UI changes. Data-Driven Framework takes this further by externalizing your test data into Excel files, allowing the same test to run with hundreds of data sets without changing a single line of code.



→ Maven Configuration

Set up and manage all project dependencies, build lifecycle, and plugin configurations using pom.xml — the standard for every professional Java project.

→ Page Object Model

Design clean, maintainable page classes that encapsulate locators and actions. Understand why POM is the industry default for Selenium projects.

→ Data-Driven Framework

Integrate Excel test data using Apache POI and run the same test scenarios with multiple data sets — exactly how enterprise QA teams operate.

Outcome: Students will learn how professional automation frameworks are created in companies and will build their own working framework as part of the program.

Testing Frameworks: JUnit, TestNG & Cucumber

No automation framework is complete without a robust test runner and reporting engine. This module covers all three major testing frameworks used in real Java Selenium projects — JUnit for unit-level testing, TestNG for advanced test management, and Cucumber for Behavior-Driven Development. Understanding all three gives you the versatility to work in any QA team, regardless of their chosen stack.

JUnit

- Annotations (@Test, @Before, @After)
- Basic Test Programs
- Assertions (assertEquals, assertTrue)
- Test Suite Configuration

TestNG

- Introduction & Installation
- Priority & Suite Configuration
- Invocation Count
- Parameterization
- Groups and Assertions
- Parallel Execution
- Rerun Failed Tests
- dependsOnMethods()

Cucumber BDD

- Introduction & Setup
- Cucumber Options
- Scenario & Scenario Outline
- Background
- Data Tables
- Tags and Hooks
- Reports and Rerun

TestNG is particularly powerful for enterprise-scale automation because it supports parallel execution across multiple browsers and environments — dramatically reducing test suite run time. Cucumber BDD bridges the gap between technical QA engineers and non-technical business stakeholders by writing test scenarios in plain English using the Gherkin language. This skill is highly valued in Agile development teams and makes you a more effective communicator on any project.

Tools & Technologies You Will Learn

Modern QA engineers are expected to work with a full ecosystem of tools beyond just Selenium and Java. This module ensures you are fluent in the complete professional toolkit that every IT company's QA team uses — from version control and CI/CD pipelines to database testing and API validation. By graduation, your resume will reflect a genuinely comprehensive skill set.

Git — Version Control

Manage your code with Git repositories. Learn branching, committing, pushing, and collaborating on shared codebases — essential for every development and QA team.

Jenkins — CI/CD Pipeline

Automate your test execution with Jenkins. Schedule test runs, integrate with Git, and generate reports automatically — the backbone of modern DevOps QA practice.

SQL & Database Testing

Write SQL queries, joins, and table operations to validate data at the database level. JDBC integration allows your Selenium scripts to connect and verify backend data directly.

API Testing Basics

Understand RESTful API concepts, HTTP methods, and API response validation. API testing knowledge is increasingly required even for UI automation roles.

Maven — Build Tool

Configure and manage Java project builds using Maven. Understand pom.xml, dependency management, and build lifecycle commands used in every enterprise Java project.

Eclipse IDE

Set up and use Eclipse as your primary development environment. Learn shortcuts, debugging tools, and project organization that maximize your coding efficiency.

IPT — Real-Time Training Program

The IPT (Integrated Practical Training) program is where everything comes together. This is not a module — it is a full real-time training experience that mirrors what you will encounter in your first week at a software company. IPT is structured in two phases, each designed to build on your existing knowledge and push you into genuinely professional-grade automation practice.

IPT Phase 1: Cucumber & Tools

- Cucumber Options & Configuration
- Scenario and Scenario Outline
- Background
- Data Table
- Tags and Hooks
- Reports and Rerun
- Git — Version Control Practice
- Jenkins — CI/CD Pipeline Setup
- SQL Queries and Database Operations
- JDBC Integration Basics
- API Testing Concepts

IPT Phase 2: Advanced Real-Time Training

- Java Real-Time Concepts
- Java 7 and Java 8 Features
- OOPs Real-Time Implementation
- Collections Real-Time Usage
- Iteration: Iterator and Enumeration
- Programs and Problem Solving

What Makes IPT Different

IPT is not lecture-based. Every session is a hands-on task or a project scenario. Trainers act as team leads, giving you assignments and reviewing your work the way a real QA manager would. By the end of IPT, you will have genuine project experience to discuss in your interviews — not just theoretical knowledge.

Selenium Advanced Topics

The advanced Selenium module prepares you for the most complex automation challenges encountered in real enterprise projects. These are the topics that separate entry-level automation testers from engineers who can walk into any QA team and contribute on day one. Every topic here is sourced directly from real automation challenges that companies face on live projects today.

1

Selenium 3 vs Selenium 4 Updates

Understand the architectural and API changes in Selenium 4 — including the new relative locators, Chrome DevTools Protocol integration, and improved grid capabilities.

2

Advanced XPath & Coordinates

Write complex XPath expressions for deeply nested, dynamic DOM structures. Use coordinate-based interactions for canvas elements and custom UI components.

3

Shadow DOM Handling

Automate modern web components that use Shadow DOM — a common challenge in enterprise web applications that basic locators cannot penetrate.

4

Chrome Options & Browser Configuration

Configure browser settings programmatically: headless mode, incognito, download paths, and security bypass options used in CI/CD environments.

5

BrowserStack Integration

Run your automation tests across real browsers and devices in the cloud using BrowserStack — the industry standard for cross-browser testing at scale.

6

Extent Report Generation

Generate beautiful, detailed HTML test reports with screenshots, logs, and pass/fail analytics that can be shared directly with QA managers and stakeholders.

ⓘ **Also Covered:** Navigation and Visibility Checks · CSS Selector Values · Broken Links Handling · Framework Development — all real-world requirements.

Manual Testing & Real-Time Industry Practice

Every great automation engineer is first a great manual tester. Understanding how to test software manually — with a sharp analytical mindset, structured test cases, and a clear understanding of what "done" looks like — is the foundation of all QA work. This module ensures you have that foundation firmly in place before and alongside your automation training.

Manual testing is not a lesser skill — it is a different and essential skill. Companies still rely heavily on manual testers, especially for exploratory testing, usability validation, and regression testing of features that are too complex or unstable to automate profitably. Understanding when to automate and when to test manually is a judgment call that experienced QA engineers make every day — and this module builds that judgment.

1 Manual Testing Real-Time Concepts

Understand SDLC, STLC, testing types, testing levels, defect lifecycle, and the documentation standards used in professional QA teams — all explained with real project examples.

2 Test Case Writing

Learn to write clear, structured, and comprehensive test cases that cover positive, negative, and boundary scenarios. Master test case templates used in industry tools like Jira and Excel.

3 Scenario-Based & Practical Testing

Work through real application testing scenarios — login modules, e-commerce flows, form validations, and more — using the same approach a QA engineer uses on a live project.

4 Live Tools & Project-Based Practice

Practice manual testing on real applications using live tools. Gain the hands-on experience that makes the difference between a candidate who talks about testing and one who has actually done it.

✅ 💡 **Practice testing like a real QA engineer with live tools and real projects — not simulated exercises.**

Start Your Career as a Software Tester

Learn Manual Testing, Java, Selenium, Frameworks, and Real-Time Projects with XpertsEdgeTechnologies — and walk out with the skills, confidence, and career support to land your first QA role.

🎯 Career Roles You Can Target

- Manual Tester
- Software Tester
- QA Engineer
- Automation Tester
- Selenium Tester
- Test Engineer
- Junior QA Analyst
- QA Automation Engineer

💼 Career Support Included

- Resume Preparation & Review
- Mock Interviews with Expert Feedback
- Common Interview Questions Practice
- Project Explanation Coaching
- Referral Support
- Ongoing Job Guidance

Our career support does not end when your training ends. We stay connected with every graduate and continue to support your job search until you are placed. Your success is our reputation.

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